

Aditya Rustagi

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 4+ years of experience designing, training, deploying, and optimizing production-grade models and large-scale data pipelines. Skilled in LLMs, agentic AI, deep learning, model inference optimization, MLOps, distributed computing, and cloud-native ML platforms

EXPERIENCE

- CVS Health** Apr 2023 - Current
 - Machine Learning Engineer*
 - Architected an enterprise multi-agent platform on Google Kubernetes Engine powering 5+ agents, integrating RAG, text-to-SQL, Palantir Foundry, and Redis. Delivered 1K+ daily insights, reduced analyst time-to-insight from hours to seconds, cut BI ticket volume by 60%, while maintaining p95 latency below 2 seconds.
 - Engineered an AI-native metrics governance platform for 200+ KPIs used by 43 teams on Azure Kubernetes Service, translating natural-language questions into deterministic SQL across Snowflake and MySQL through custom MCP, MetricFlow, and Data Junction
 - Optimized critical batch-processing pipelines operating on more than 3 billion records weekly, reducing compute time by 98% and generating \$2.4M in annual infrastructure savings
 - Implemented MLOps controls for model-drift detection, automated retraining, fault-tolerant error handling, centralized observability with Prometheus, Grafana, Loki, and Tempo, reducing manual intervention by 33%
 - Lead a three-person team on a personal proposed agentic SDLC compliance reviewer using RAG backed rule retrieval, webhooks, and confidence scoring to automate pull-request reviews against enterprise coding standards
- NVIDIA** June 2022 - Sept 2022
 - Software Engineering Intern - RAPIDS Data Science*
 - Developed CUDA-accelerated data-science kernels in the RAPIDS ecosystem, delivering up to 6x performance improvements on large-scale tabular workloads. Optimized HoloViews rendering operations to double small-dataset visualization throughput and reduce latency
 - Contributed optimizations to open-source LUX and RAPIDS libraries, advancing GPU-native data visualization
- CISCO** Jan 2021 - Jul 2021
 - Software Engineering Intern*
 - Built an automated ETL pipeline using Python, Selenium, clustering based transformations, and SQL to extract and normalize 1M+ gateway-communication records, reducing manual data processing effort by 77% and enabling engineers to identify performance bottlenecks
- Rootee Health** Apr 2020 - Jan 2021
 - Machine Learning Engineer*
 - Developed and deployed TensorRT optimized TensorFlow models for real-time object tracking and image segmentation on edge devices, improving inference speed by 2x for medical-diagnostic applications
 - Built an anomaly detection pipeline that prevented 92% of faulty images from entering downstream workflows
 - Deployed the startup's flagship ocular-disease diagnostic pipeline and supported investor demonstrations preceding the company's acquisition

PROJECTS

- AI Document Assistant: LangGraph-Orchestrated RAG Pipeline:**
 - Built LLM-powered document assistant with LangGraph orchestration, implementing semantic search using ChromaDB embeddings and intelligent tool routing for question answering on any given document
 - Engineered RAG pipeline using FastAPI, HuggingFace embeddings, and OpenAI GPT-3.5-turbo with optimized text chunking and persistent vector storage for real-time document analysis

EDUCATION

- Master of Science in Machine Learning and Data Science**
 - University of California San Diego*
- Dual Degree - B.E in Electrical and Electronics & M.Sc in Chemistry**
 - Birla Institute of Technology and Science, Pilani*

SKILLS SUMMARY

Languages:	Python, SQL, R, C++, CUDA, GO, MATLAB
Machine Learning:	PyTorch, TensorFlow, scikit-learn, Hugging Face, TensorRT, RAG, LangChain, LangGraph
Computation:	Spark, Snowflake, Palantir Foundry, Pandas, NumPy, cuDF, Redis, Dask
CI/CD Tools:	Vertex AI, GCP, Azure, Kubernetes, Kubeflow, Airflow, Jenkins, Docker, Github, ArgoCD